Laplace Transform

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Laplace Transform

Let $f : [0, \infty) \to \mathbb{R}$ be a function. If $f(t)$ satisfies the following Conditions:

- $f(t)$ is piecewise continuous on the interval $[0, b]$ for each $b > 0$
- There exist real constants $M \geq 0$ and real positive number a such that

$$|e^{-at} f(t)| \leq M, \quad \text{i.e.,} \, |f(t)| \leq Me^{at} \quad \text{for all } t > 0,$$

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Definition(Laplace Transform)

Then, the Laplace transform of $f(t)$, denoted by $\mathcal{L}\{f(t)\}$ or $F(s)$, is defined by

$$\mathcal{L}\{f(t)\} = F(s) = \int_0^\infty e^{-st} f(t) \, dt,$$

provided this integral exists. Here s is a positive real number or a complex number.

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Observe

$$\left| \int_0^\infty e^{-st} f(t)\, dt \right| \leq \int_0^\infty |f(t)| e^{-st} dt \leq M \int_0^\infty e^{-(s-a)t} dt = \frac{M}{s-a}, \quad s > a$$

Note

The variable s also takes complex values, but for the time being, assume s to be real.

Example

Compute $\mathcal{L}\{1\}$.

Example

Compute $\mathcal{L}\{1\}$.

Solution.

By definition

$$\mathcal{L}\{1\} = \int_0^\infty \mathbf{e}^{-st}\, dt = \left[\frac{e^{-st}}{-s}\right]_0^\infty = -\frac{1}{-s} \qquad \textit{provided } -s < 0$$

Or, with some simplification we have,

$$\mathcal{L}\{1\} = \frac{1}{s} \qquad \textit{provided } s > 0$$

Example

Compute $\mathcal{L}\{e^{at}\}$

Example

Compute $\mathcal{L}\left\{\mathbf{e}^{a\,t}\right\}$

Solution.

Plugging the function into the definition of the transform and do a little simplification.

$$\begin{aligned}
\mathcal{L}\left\{\mathbf{e}^{a\,t}\right\} &= \int_0^\infty \mathbf{e}^{-s\,t}\mathbf{e}^{a\,t}\,dt \\
&= \int_0^\infty \mathbf{e}^{(a-s)t}\,dt \\
&= \left[\frac{\mathbf{e}^{(a-s)t}}{a-s}\right]_0^\infty \\
&= -\frac{1}{a-s} \qquad \textit{provided } a-s<0 \\
&= \frac{1}{s-a} \qquad \textit{provided } s>a
\end{aligned}$$

Example

Compute $\mathcal{L}\{\sin(at)\}$.

Example

Compute $\mathcal{L}\{\sin(at)\}$.

Solution.

Plugging the function into the definition of the transform

$$\begin{aligned}
\mathcal{L}\{\sin(at)\} &= \int_0^\infty \mathbf{e}^{-st}\sin(at)\,dt \\
&= -\left(\frac{1}{a}\mathbf{e}^{-st}\cos(at)\right)\Big|_0^\infty - \frac{s}{a}\int_0^\infty \mathbf{e}^{-st}\cos(at)\,dt \\
&= \frac{1}{a} - \frac{s}{a}\left(\frac{1}{a}\mathbf{e}^{-st}\sin(at)\Big|_0^\infty + \frac{s}{a}\int_0^\infty \mathbf{e}^{-st}\sin(at)\,dt\right) \quad \textit{provided } s>0 \\
&= \frac{1}{a} - \frac{s^2}{a^2}\int_0^\infty e^{-st}\sin(at)dt \qquad \textit{provided } s>0
\end{aligned}$$

$$\implies \mathcal{L}\{\sin(at)\} = \frac{1}{a} - \frac{s^2}{a^2}\mathcal{L}\{\sin(at)\}$$

$$\implies \mathcal{L}\{\sin(at)\} = \frac{a}{s^2+a^2} \qquad \textit{provided } s>0$$

Derivative property of Laplace transform

Suppose a differentiable function $f(t)$ has a Laplace transform $F(s)$,

We can obtain the Laplace transform of it's derivative $f'(t)$:

$$\mathcal{L}\{f'(t)\} = \int_0^\infty e^{-st} f'(t)dt.$$

Similarly, if the function $f(t)$ is twice, thrice,..., n-times differentiable,

We can obtain the Laplace transforms of $f''(t)$, $f'''(t), \ldots, f^{(n)}(t)$ respectively.

The following theorems relate to the Laplace transforms of the derivatives of a function

Note

These results are extremely helpful in solving differential equations of different orders.

Derivative property of Laplace transform

Theorem.

If $f(t)$ is a differentiable function of t and $\mathcal{L}\{f(t)\} = F(s)$, then

$$\mathcal{L}\{f'(t)\} = sF(s) - f(0)$$

where $f(0)$ is the value of $f(t)$ at $t = 0$.

Derivative property of Laplace transform

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Theorem.

If $f(t)$ is a twice differentiable function of t and $\mathcal{L}\{f(t)\} = F(s)$, then

$$\mathcal{L}\{f''(t)\} = s^2F(s) - sf(0) - f'(0)$$

where $f(0)$ and $f'(0)$ are the values of $f(t)$ and $f'(t)$, respectively at $t = 0$.

We can generalize above results as follows:

Derivative property of Laplace transform

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We can generalize above results as follows:

Theorem.

If $f(t)$ is an n-times differentiable function of t, and $\mathcal{L}\{f(t)\} = F(s)$, then

$$\mathcal{L}\{f^{(n)}(t)\} = s^nF(s) - s^{n-1}f(0) - s^{n-2}f'(0) - \cdots - sf^{(n-2)}(0) - f^{(n-1)}(0)$$

Example

Consider the following first order ODE

$$\frac{dx}{dt} + 3x = 0, \quad x(0) = 1$$

Find $\mathcal{L}\{x\}$.

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Consider the following first order ODE

$$\frac{dx}{dt} + 3x = 0, \quad x(0) = 1$$

Find $\mathcal{L}\{x\}$.

Solution.

Given differential equation is

$$\frac{dx}{dt} + 3x = 0$$

By taking Laplace transform on both sides of the equation ,

$$\mathcal{L}\left\{\frac{dx}{dt}\right\} + \mathcal{L}\{x\} = 0$$

$$\implies s\mathcal{L}\{x\} - x(0) + 3\mathcal{L}\{x\} = 0$$

$$\implies (s+3)\mathcal{L}\{x\} = 1$$

$$\implies \mathcal{L}\{x\} = \frac{1}{s+3}$$

Laplace transform of some elementary functions

- $\mathcal{L}\{1\} = \dfrac{1}{s}, \quad s > 0$
- $\mathcal{L}\{t\} = \dfrac{1}{s^2}, \quad s > 0$
- $\mathcal{L}\{t^n\} = \dfrac{n!}{s^{n+1}}$, n an integer or

  $\mathcal{L}\{t^n\} = \dfrac{\Gamma(n+1)}{s^{n+1}}, \quad n+1 > 0, \quad n$ not an integer
- $\mathcal{L}\{e^{at}\} = \dfrac{1}{s-a}, \quad s > a$
- $\mathcal{L}\{\sin at\} = \dfrac{a}{s^2+a^2}, \quad s > 0$
- $\mathcal{L}\{\cos at\} = \dfrac{s}{s^2+a^2}, \quad s > 0$
- $\mathcal{L}\{\sinh at\} = \dfrac{a}{s^2-a^2}, \quad s > |a|\}$
- $\mathcal{L}\{\cosh at\} = \dfrac{s}{s^2-a^2}, \quad s > |a|$

Elementary Properties of Laplace transform

The Laplace transform has many interesting and useful properties, the most fundamental of which is linearity property.

Theorem (Linearity Property).

If $f_1(t)$ and $f_2(t)$ are two functions whose Laplace transforms exist, then

$$\mathcal{L}\{af_1(t) + bf_2(t)\} = a\mathcal{L}\{f_1(t)\} + b\mathcal{L}\{f_2(t)\}$$

where a and b are arbitrary constants.

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where a and b are arbitrary constants.

Theorem (First Shifting Theorem).

If $f(t)$ is a Laplace transformable function, i.e., $\mathcal{L}\{f(t)\} = F(s)$, then

$$\mathcal{L}\{e^{at}f(t)\} = F(s-a)$$

Elementary Properties of Laplace transform Cont...

Definition (Heaviside unit function)

The Heaviside unit function is defined as,

$$u_a(t) = \begin{cases} 1 & t \geq a \\ 0 & t < a \end{cases}$$

Elementary Properties of Laplace transform Cont...

Definition (Heaviside unit function)

The Heaviside unit function is defined as,

$$u_a(t) = \begin{cases} 1 & t \geq a \\ 0 & t < a \end{cases}$$

Theorem (Second Shifting Theorem).

If $f(t)$ is a Laplace transformable function, i.e., $\mathcal{L}\{f(t)\} = F(s)$ and

$$g(t) = u_a(t)f(t-a) = \begin{cases} f(t-a), & t \geq a \\ 0, & t < a \end{cases}$$

then

$$\mathcal{L}\{g(t)\} = \mathcal{L}\{u_a(t)f(t-a)\} = e^{-as}F(s).$$

Elementary Properties of Laplace transform Cont...

Theorem (Change of Scale Theorem).

If $\mathcal{L}\{f(t)\} = F(s)$, *then*

$$\mathcal{L}\{f(at)\} = \frac{1}{a} F\left(\frac{s}{a}\right), \quad a > 0$$

Elementary Properties of Laplace transform Cont...

Theorem (Change of Scale Theorem).

If $\mathcal{L}\{f(t)\} = F(s)$, *then*

$$\mathcal{L}\{f(at)\} = \frac{1}{a} F\left(\frac{s}{a}\right), \quad a > 0$$

Theorem (Multiplication by t).

If $\mathcal{L}\{f(t)\} = F(s)$, *then*

$$\mathcal{L}\{tf(t)\} = -\frac{d}{ds}F(s)$$

and in general

$$\mathcal{L}\{t^n f(t)\} = (-1)^n \frac{d^n}{ds^n}F(s).$$

Elementary Properties of Laplace transform Cont...

Theorem (Laplace of Integration).

If $\mathcal{L}\{f(t)\} = F(s)$, *then*

$$\mathcal{L}\left\{\int_0^t f(u)\, du\right\} = \frac{F(s)}{s}$$

Elementary Properties of Laplace transform Cont...

Theorem (Laplace of Integration).

If $\mathcal{L}\{f(t)\} = F(s)$, *then*

$$\mathcal{L}\left\{\int_0^t f(u)\, du\right\} = \frac{F(s)}{s}$$

Theorem (Division by t).

If $\mathcal{L}\{f(t)\} = F(s)$, *then*

$$\mathcal{L}\left\{\frac{f(t)}{t}\right\} = \int_s^\infty F(u)\, du$$

assuming that $\mathcal{L}\left\{\frac{f(t)}{t}\right\} \to 0$ *as* $s \to \infty$.

Example

Find the Laplace transforms of the given functions.

(a) $f(t) = 6\mathbf{e}^{-5t} + \mathbf{e}^{3t} + 5t^3 - 9$

(b) $g(t) = 4\cos(4t) - 9\sin(4t) + 2\cos(10t)$

(c) $h(t) = 3\sinh(2t) + 3\sin(2t)$

(d) $g(t) = \mathbf{e}^{3t} + \cos(6t) - \mathbf{e}^{3t}\cos(6t)$

(e) $f(t) = 5t^3 + t\sin(t) - e^{2t}\cos(6t)$

Solution.

(a)

$$F(s) = 6\frac{1}{s-(-5)} + \frac{1}{s-3} + 5\frac{3!}{s^{3+1}} - 9\frac{1}{s}$$
$$= \frac{6}{s+5} + \frac{1}{s-3} + \frac{30}{s^4} - \frac{9}{s}$$

(b)

$$G(s) = 4\frac{s}{s^2+(4)^2} - 9\frac{4}{s^2+(4)^2} + 2\frac{s}{s^2+(10)^2}$$
$$= \frac{4s}{s^2+16} - \frac{36}{s^2+16} + \frac{2s}{s^2+100}$$

(c)

$$H(s) = 3\frac{2}{s^2-(2)^2} + 3\frac{2}{s^2+(2)^2}$$
$$= \frac{6}{s^2-4} + \frac{6}{s^2+4}$$

(d)

$$G(s) = \frac{1}{s-3} + \frac{s}{s^2+(6)^2} - \frac{s-3}{(s-3)^2+(6)^2}$$
$$= \frac{1}{s-3} + \frac{s}{s^2+36} - \frac{s-3}{(s-3)^2+36}$$

Solution cont...

(e) We know that

$$\mathcal{L}\{t^3\} = \frac{3!}{s^4} = \frac{6}{s^4}$$

and

$$\mathcal{L}\{\sin t\} = \frac{1}{s^2+1}$$

$$\implies \mathcal{L}\{t\sin t\} = -\frac{d}{ds}\left(\frac{1}{s^2+1}\right) \quad \text{[Multiplication by } t \text{ property]}$$

$$= \frac{2s}{(s^2+1)^2}$$

and

$$\mathcal{L}\{\cos 6t\} = \frac{s}{s^2+6^2}$$

$$\implies \mathcal{L}\{e^{2t}\cos 6t\} = \frac{s-2}{(s-2)^2+36} \quad \text{[First shifting theorem]}$$

Therefore

$$\mathcal{L}\{f(t)\} = 5\mathcal{L}\{t^3\} + \mathcal{L}\{t\sin t\} - \mathcal{L}\{e^{2t}\cos 6t\} \quad \text{[Linearity property]}$$

$$= \frac{30}{s^4} + \frac{2s}{s^2+1} - \frac{s-2}{(s-2)^2+36}$$

Example

Find the transform of each of the following functions.

(a) $f(t) = t\cosh(3t)$

(b) $h(t) = t^2 \sin(2t)$

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Solution.

(a)

$$\mathcal{L}\{t\cosh(3t)\} = -\frac{d}{ds}\mathcal{L}\{\cosh(3t)\} = -\frac{d}{ds}\left(\frac{s}{s^2-a^2}\right)$$
$$= \frac{s^2+9}{(s^2-9)^2}$$

(b)

$$\mathcal{L}\{t^2\sin(2t)\} = (-1)^2\frac{d^2}{ds^2}\mathcal{L}\{\sin(2t)\} = \frac{d^2}{ds^2}\left(\frac{3}{s^2+9}\right)$$
$$= \frac{12s^2-16}{(s^2+4)^3}$$

Example

Find the Laplace transforms of each of the following functions

(a) $f(t) = \dfrac{\sin t}{t}$ **Ans.** $\dfrac{\pi}{2} - \tan^{-1} s$

(b) $g(t) = \dfrac{1 - \cos 2t}{t}$ **Ans.** $\ln \dfrac{\sqrt{s^2+4}}{s}$

(c) $h(t) = \dfrac{e^{4t} - e^{-3t}}{t}$ **Ans.** $\ln \dfrac{s+3}{s-4}$

(d) $f(t) = \dfrac{\sin^2 t}{t}$ **Ans.** $\frac{1}{4} \ln \left(\dfrac{s^2+4}{s^2} \right)$

(e) $g(t) = \dfrac{\cos 4t - \cos 5t}{t}$ **Ans.** $\frac{1}{2} \ln \dfrac{s^2+25}{s^2+16}$

(f) $h(t) = \dfrac{1 - e^t}{t}$ **Ans.** $\ln \dfrac{s-1}{s}$

(g) $f(t) = \dfrac{e^{-t} \sin t}{t}$ **Ans.** $\dfrac{\pi}{2} - \tan^{-1}(s+1)$

Solution.

(a) *We know that*

$$\mathcal{L}\{\sin t\} = \frac{1}{s^2+1}$$

Therefore

$$\mathcal{L}\left\{\frac{\sin t}{t}\right\} = \int_s^\infty \frac{1}{u^2+1}du = \tan^{-1} u\Big|_s^\infty = \frac{\pi}{2} - \tan^{-1} s$$

(b) *We have*

$$\mathcal{L}\{1-\cos 2t\} = \mathcal{L}\{1\} - \mathcal{L}\{\cos 2t\} = \frac{1}{s} - \frac{s}{s^2+4}$$

Therefore

$$\mathcal{L}\left\{\frac{1-\cos 2t}{t}\right\} = \int_s^\infty \left(\frac{1}{u} - \frac{u}{u^2+4}\right) du = (\ln u - \ln\sqrt{u^2+4})\Big|_s^\infty = \ln\frac{\sqrt{s^2+4}}{s}$$

(c) *We have*

$$\mathcal{L}\{e^{4t} - e^{-3t}\} = \mathcal{L}\{e^{4t}\} - \mathcal{L}\{e^{-3t}\} = \frac{1}{s-4} - \frac{1}{s+3}$$

Therefore

$$\mathcal{L}\left\{\frac{e^{4t}-e^{-3t}}{t}\right\} = \int_s^\infty \left(\frac{1}{u-4} - \frac{1}{u+3}\right) du = \ln(u-4) - \ln(u+3)\Big|_s^\infty = \ln\frac{s+3}{s-4}$$

Solution cont. . .

(g) We know that

$$\mathcal{L}\{\sin t\} = \frac{1}{1+s^2}$$

$$\implies \mathcal{L}\{e^{-t}\sin t\} = \frac{1}{(s+1)^2+1} \qquad \text{[First shitying theorem]}$$

$$\implies \mathcal{L}\left\{\frac{e^{-t}\sin t}{t}\right\} = \int_s^\infty \frac{1}{(u+1)^2+1}\,du \qquad \text{[Division by } t \text{ property]}$$

$$\implies \mathcal{L}\left\{\frac{e^{-t}\sin t}{t}\right\} = \tan^{-1} u\Big|_s^\infty$$

$$\implies \mathcal{L}\left\{\frac{e^{-t}\sin t}{t}\right\} = \frac{\pi}{2} - \tan^{-1}(s+1)$$

Example

Find the Laplace transform of each of the following functions

(a) $f(t) = \begin{cases} \sin(t-1) & t \geq 1 \\ 0 & t < 1 \end{cases}$ **Ans.** $\dfrac{e^{-s}}{s^2+1}$

(b) $g(t) = \begin{cases} (t-1)\sin(t-1) & t \geq 1 \\ 0 & t < 1 \end{cases}$ **Ans.** $\dfrac{2se^{-s}}{(s^2+1)^2}$

(c) $h(t) = \sin t \; u_\pi(t)$ **Ans.** $-\dfrac{e^{-\pi s}}{s^2+1}$

(d) $f(t) = (t-1)^2 u_1(t)$ **Ans.** $\dfrac{2e^{-s}}{s^2}$

(e) $g(t) = (t-1)^2 u_2(t)$ **Ans.** $e^{-2s}\left(\dfrac{2}{s^3}+\dfrac{2}{s^2}+\dfrac{1}{s}\right)$

(f) $h(t) = 4u_\pi \cos t$ **Ans.** $e^{-\pi s}\left(\frac{-4s}{s^2+1}\right)$

(g) $f(t) = 9u_\pi \sin t$ **Ans.** $e^{-\pi s}\left(\frac{-9}{s^2+1}\right)$

(h) $g(t) = u_2 e^{-3t}$ **Ans.** $e^{-2s}\left(\frac{e^{-6}}{s+3}\right)$

Solution.

(a) *We have* $\mathcal{L}\{\sin t\} = \dfrac{1}{s^2+1}$. *Therefore, by second shifting theorem*

$$\mathcal{L}\{f(t)\} = \mathcal{L}\{u_1(t)\sin(t-1)\} = e^{-s}\frac{1}{s^2+1} = \frac{e^{-s}}{s^2+1}$$

(b) *We have* $\mathcal{L}\{\sin t\} = \dfrac{1}{s^2+1} \implies \mathcal{L}\{t\sin t\} = -\dfrac{d}{ds}\left(\dfrac{1}{s^2+1}\right) = \dfrac{2s}{(s^2+1)^2}$.

Therefore, by second shifting theorem

$$\mathcal{L}\{g(t)\} = \mathcal{L}\{u_1(t)(t-1)\sin(t-1)\} = e^{-s}\frac{2s}{(s^2+1)^2} = \frac{2se^{-s}}{(s^2+1)^2}$$

(c) *We have*

$$u_\pi(t)\sin t = u_\pi(t)\sin(t-\pi+\pi) = -u_\pi(t)\sin(t-\pi)$$

$$\implies \mathcal{L}\{u_\pi(t)\sin t\} = -\mathcal{L}\{u_\pi(t)\sin(t-\pi)\} = -e^{-\pi s}\frac{1}{s^2+1} = -\frac{e^{-\pi s}}{s^2+1}$$

(d)

$$\mathcal{L}\{(t-1)^2 u_1(t)\} = e^{-s}\mathcal{L}\{t^2\} = \frac{2e^{-s}}{s^2}$$

Solution cont. . .

(h) We have

$$g(t) = u_2 e^{-3t} = u_2 e^{-3(t-2+2)} = e^{-6} u_2 e^{-3(t-2)}$$

$$\begin{aligned} \Longrightarrow \quad \mathcal{L}\{g(t)\} &= e^{-6}\mathcal{L}\{u_2 e^{-3(t-2)}\} && \text{[Linearity property]} \\ &= e^{-6} e^{-2s}\mathcal{L}\{e^{-3t}\} && \text{[2nd shifting theorem]} \\ &= e^{-6} e^{-2s}\frac{1}{s+3} \end{aligned}$$

$$\Longrightarrow \quad \mathcal{L}\{g(t)\} = e^{-2s}\left(\frac{e^{-6}}{s+3}\right)$$

Example

Let $f(t) = \cos(t^2)$ and $g(t) = \sin(t^2)$ defined in the interval $[0, \infty)$. Suppose that the Laplace transform of $f(t)$ and $g(t)$ are given by $F(s)$ and $G(s)$, respectively, both defined and differentiable for $s > 0$. Then prove the following:

(a) $sF(s) - 2G'(s) = 1$.

(b) $sG(s) + 2F'(s) = 0$.

Solution

Given that $f(t) = \cos(t^2), g(t) = \sin(t^2)$ and $\mathcal{L}\{f(t)\} = F(s), \mathcal{L}\{g(t)\} = G(s)$.

(a)

$$\mathcal{L}\{f'(t)\} = sF(s) - f(0) \qquad \text{[Derivative property]}$$
$$\implies \mathcal{L}\{-2t\sin t^2\} = sF(s) - 1$$
$$\implies -2\mathcal{L}\{t\sin t^2\} = sF(s) - 1 \qquad \text{[Linearity property]}$$
$$\implies -2\left(-\frac{d}{ds}\mathcal{L}\{\sin t^2\}\right) = sF(s) - 1 \qquad \text{[Multiplication by } t \text{ property]}$$
$$\implies 2G'(s) = sF(s) - 1$$
$$\implies sF(s) - 2G'(s) = 1$$

(b)

$$\mathcal{L}\{g'(t)\} = sG(s) - g(0) \qquad \text{[Derivative property]}$$
$$\implies \mathcal{L}\{2t\cos t^2\} = sG(s)$$
$$\implies 2\mathcal{L}\{t\cos t^2\} = sG(s) \qquad \text{[Linearity property]}$$
$$\implies 2\left(-\frac{d}{ds}\mathcal{L}\{\cos t^2\}\right) = sG(s) \qquad \text{[Multiplication by } t \text{ property]}$$
$$\implies -2F'(s) = sG(s)$$
$$\implies sG(s) + 2F'(s) = 0$$

Exercise 1:

Find the Laplace transforms of the given functions.

(a) $f(t) = 6\mathbf{e}^{-5t} + \mathbf{e}^{3t} + 5t^3 - 9$

Ans. $F(s) = \frac{6}{s+5} + \frac{1}{s-3} + \frac{30}{s^4} - \frac{9}{s}$

(b) $g(t) =$
$4\cos(4t) - 9\sin(4t) + 2\cos(10t)$

Ans. $G(s) = \frac{4s}{s^2+16} - \frac{36}{s^2+16} + \frac{2s}{s^2+100}$

(c) $h(t) = 3\sinh(2t) + 3\sin(2t)$

Ans. $H(s) = \frac{6}{s^2-4} + \frac{6}{s^2+4}$

(d) $g(t) = \mathbf{e}^{3t} + \cos(6t) - \mathbf{e}^{3t}\cos(6t)$

Ans. $G(s) = \frac{1}{s-3} + \frac{s}{s^2+36} - \frac{s-3}{(s-3)^2+36}$

Exercise 2:

Find the transform of each of the following functions.

(a) $f(t) = t\cosh(3t)$

Ans. $F(s) = \frac{s^2+9}{(s^2-9)^2}$

(b) $h(t) = t^2\sin(2t)$

Ans. $H(s) = \frac{12s^2-16}{(s^2+4)^3}$

Exercise 3:

Find the Laplace transform of each function below.

(a) $f(t) = t^3 - t^2 + 5t + 2$

Ans. $F(s) = \frac{2s^3+5s^2-2s+6}{s^4}$

(b) $f(t) = -2\cos 6t + 5\sin 6t$

Ans. $F(s) = \frac{-2s+30}{s^2+36}$

(c) $f(t) = 3e^t - 4e^{2t} + 2e^{-4t}$

Ans. $F(s) = \frac{s^2-12s-4}{(s-1)(s-2)(s+4)}$

(d) $f(t) = 7t + 6e^t - 2e^{-t} - 10$

Ans. $F(s) = \frac{-6s^3+15s^2+10s-7}{s^4-s^2}$

(e) $f(t) = \sin 2t \sin 3t$

Ans. $F(s) = \frac{s}{2(s^2+1)} - \frac{s}{2(s^2+25)}$

(f) $f(t) = \sin 2t \cos 2t$

Ans. $F(s) = \frac{2}{s^2+16}$

(g) $f(t) = \cos^2 5t$

Ans. $F(s) = \frac{1}{2}\left[\frac{1}{s} + \frac{s}{s^2+100}\right]$

(h) $f(t) = t\cos^2 5t$

Ans. $F(s) = \frac{1}{2}\left[\frac{1}{s^2} + \frac{s^2-100}{(s^2+100)^2}\right]$

(i) $f(t) = te^{at}\cos bt$

Ans. $F(s) = \frac{(s-a)^2-b^2}{((s-a)^2+b^2)^2}$

(j) $f(t) = te^{at}\sin bt$

Ans. $F(s) = \frac{2b(s-a)}{[(s-a)^2+b^2]^2}$

(k) $f(t) = t^3\cos 2t$

Ans. $F(s) = \frac{6s^4-144s^2+96}{(s^2+4)^4}$

(l) $f(t) = \cos(\alpha t + \beta)$

Ans. $F(s) = \frac{s\cos\beta-\alpha\sin\beta}{s^2+\alpha^2}$

(m) $f(t) = \sin(\alpha t + \beta)$

Ans. $F(s) = \frac{s\sin\beta+\alpha\cos\beta}{s^2+\alpha^2}$

(n) $f(t) = f(t) = \sin^3 2t$

Ans. $F(s) = \frac{48}{s^4+40s^2+144}$

Exercise 4:

Find the Laplace transform of each of the following.

- $g(t) = 10u_{12}(t) + 2(t-6)^3 u_6(t) - \left(7 - \mathbf{e}^{12-3t}\right) u_4(t)$.

  Ans. $G(s)\ \frac{10\mathbf{e}^{-12s}}{s} + \frac{12\mathbf{e}^{-6s}}{s^{3+1}} - \left(\frac{7}{s} - \frac{1}{s+3}\right)\mathbf{e}^{-4s}$.

- $f(t) = -t^2 u_3(t) + \cos(t)\, u_5(t)$.

  Ans. $F(s) = -\left(\frac{2}{s^3} + \frac{6}{s^2} + \frac{9}{s}\right)\mathbf{e}^{-3s} + \left(\frac{s\cos(5) - \sin(5)}{s^2+1}\right)\mathbf{e}^{-5s}$.

- $h(t) = \begin{cases} t^4 & \text{if } t < 5 \\ t^4 + 3\sin\left(\frac{t}{10} - \frac{1}{2}\right) & \text{if } t \geq 5 \end{cases}$

  Ans. $H(s) = \frac{24}{s^5} + \frac{\frac{3}{10}\mathbf{e}^{-5s}}{s^2 + \frac{1}{100}}$.

- $f(t) = \begin{cases} t & \text{if } t < 6 \\ -8 + (t-6)^2 & \text{if } t \geq 6 \end{cases}$

  Ans. $F(s) = \frac{1}{s^2} + \left(\frac{2}{s^3} - \frac{1}{s^2} - \frac{14}{s}\right)\mathbf{e}^{-6s}$

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Laplace transform of periodic functions

Let $f(t)$ be a periodic function with period T, i.e., $f(t+T) = f(t)$, then, from the definition of Laplace transform, we have

$$\mathcal{L}\{f(t)\} = F(s) = \int_0^\infty f(t)e^{-st}dt$$

The above expression can also be written as,

$$F(s) = \int_0^T f(t)e^{-st}dt + \int_T^{2T} f(t)e^{-st}dt + \int_{2T}^{3T} f(t)e^{-st}dt + \cdots$$

In the second integral put t = u + T, in the third integral put t = u + 2T, and so on. Then

$$F(s) = \int_0^T f(t)\,e^{-st}dt + \int_0^T f(u+T)\,e^{-s(u+T)}dt + \int_0^T f(u+2T)\,e^{-s(u+2T)}dt + \cdots$$

$$= \int_0^T f(t)\,e^{-st}dt + e^{-sT}\int_0^T f(u+T)\,e^{-su}dt + e^{-2sT}\int_0^T f(u+2T)\,e^{-su}dt + \cdot$$

As the function $f(t)$ is a periodic function, therefore,

As the function $f(t)$ is a periodic function, therefore,

$$f(t) = f(t+T) = f(t+2T) = \cdots$$

Thus,

$$F(s) = \int_0^T f(t)\, e^{-st} dt + e^{-sT} \int_0^T f(u) e^{-su} du + e^{-2sT} \int_0^T f(u) e^{-su} du + \cdots$$

$$= \left[1 + e^{-sT} + e^{-2sT} + \cdots\right] \int_0^T f(t)\, e^{-st} dt$$

$$= \frac{1}{1 - e^{-sT}} \int_0^T f(t)\, e^{-st} dt$$

Thus,

$$\boxed{\mathcal{L}\{f(t)\} = \frac{\int_0^T f(t)\, e^{-st} dt}{1 - e^{-sT}}}$$

Example

Find the Laplace transform of function $f(t) = \sin \pi t$ using periodicity property of Laplace transform.

Example

Find the Laplace transform of function $f(t) = \sin \pi t$ using periodicity property of Laplace transform.

Solution.

Given $f(t) = \sin \pi t$ *and clearly,* $T = 2$. *Therefore*

$$\mathcal{L}\{f(t)\} = \frac{\int_0^2 \sin \pi t \; e^{-st} dt}{1 - e^{-2s}}$$

$$= \frac{1}{[1 - e^{-2s}]} \left[\frac{\pi \left(1 - e^{-2s}\right)}{s^2 + \pi^2} \right]$$

$$= \frac{\pi}{s^2 + \pi^2}$$

Example

Find the Laplace transform of the periodic function $f(t) = t^2, \quad f(t+2) = f(t)$

Example

Find the Laplace transform of the periodic function $f(t) = t^2, \quad f(t+2) = f(t)$

Solution.

Here $f(t) = t^2$ *and* $T = 2$. *Thus,*

$$\mathcal{L}\{f(t)\} = \frac{\int_0^T f(t)\, e^{-st}\, dt}{1 - e^{-sT}} = \frac{1}{1 - e^{-2s}} \int_0^2 t^2 e^{-st}\, dt$$

$$= \frac{1}{1 - e^{-2s}} \left[t^2 \frac{e^{-st}}{-s} \Big|_0^2 + \frac{2}{s} \int_0^2 t e^{-st}\, dt \right]$$

$$= \frac{1}{1 - e^{-2s}} \left[-\frac{4e^{-2s}}{s} + \frac{2}{s} \left(\frac{te^{-st}}{-s} \Big|_0^2 + \frac{1}{s} \int_0^1 e^{-st}\, dt \right) \right]$$

$$= \frac{1}{1 - e^{-2s}} \left[-\frac{4e^{-2s}}{s} + \frac{2}{s} \left(-\frac{2e^{-2s}}{s} - \frac{e^{-st}}{s^2} \Big|_0^2 \right) \right]$$

$$= \frac{1}{1 - e^{-2s}} \left[\frac{2}{s^3} - \left(\frac{4}{s} + \frac{4}{s^2} + \frac{2}{s^3} \right) e^{-2s} \right]$$

Example

Find the Laplace transform of the following periodic function

$$f(t) = \begin{cases} t & 0 \leq t < 1 \\ 2 - t & 1 \leq t < 2 \end{cases} \qquad f(t+2) = f(t)$$

Example

Find the Laplace transform of the following periodic function

$$f(t) = \begin{cases} t & 0 \le t < 1 \\ 2-t & 1 \le t < 2 \end{cases} \qquad f(t+2) = f(t)$$

Solution.

Here $f(t)$ is given to be a periodic function of period 2π. Thus,

$$\mathcal{L}\{f(t)\} = \frac{1}{1-e^{-2s}} \int_0^2 f(t)e^{-st}dt = \frac{1}{1-e^{-2s}} \left[\int_0^1 te^{-st}dt + \int_1^2 (2-t)e^{-st}dt \right]$$

$$= \frac{1}{1-e^{-2s}} \left[\left(-\frac{te^{-st}}{s} - \frac{e^{-st}}{s^2} \right) \Big|_0^1 + \left(-\frac{(2-t)e^{-st}}{s} + \frac{e^{-st}}{s^2} \right) \Big|_1^2 \right]$$

$$= \frac{1}{1-e^{-2s}} \left[-\frac{e^{-s}}{s} - \frac{e^{-s}}{s^2} + \frac{1}{s^2} + \frac{e^{-2s}}{s^2} + \frac{e^{-s}}{s} - \frac{e^{-s}}{s^2} \right]$$

$$= \frac{1}{1-e^{-2s}} \left[\frac{1}{s^2} + \frac{e^{-2s}}{s^2} - 2\frac{e^{-s}}{s^2} \right] = \frac{(1-e^{-s})^2}{s^2(1-e^{-2s})}$$

Exercise 5:

Find the Laplace transform of the following periodic function

(a) $f(t) = t^2, \quad f(t+2) = f(t).$

Ans. $\mathcal{L}\{f(t)\} = \frac{1}{1-e^{-2s}}\left[\frac{2}{s^3} - \left(\frac{4}{s} + \frac{4}{s^2} + \frac{2}{s^3}\right)e^{-2s}\right]$

(b) $f(t) = \begin{cases} t & \text{if } 0 \le t < 1 \\ 2-t & \text{if } 1 \le t < 2 \end{cases} \qquad f(t+2) = f(t)$ **Ans.** $\mathcal{L}\{f(t)\} = \frac{(1-e^{-s})^2}{s^2(1-e^{-2s})}$

(c) $f(t) = \begin{cases} 1 & \text{if } 0 \le t < 1 \\ 0 & \text{if } 1 \le t < 2 \end{cases} \qquad f(t+2) = f(t)$ **Ans.** $\mathcal{L}\{f(t)\} = \frac{1}{s(1+e^{-s})}$

(d) $f(t) = \begin{cases} t^2 & \text{if } 0 \le t < 2 \\ 2 & \text{if } 2 \le t < 3 \end{cases} \qquad f(t+3) = f(t).$

Ans. $\mathcal{L}\{f(t)\} = \frac{1}{1+e^{-3s}}\left[\frac{2}{s^3} - e^{-2S}\left(\frac{2}{s^3} + \frac{4}{s^2} + \frac{2}{s}\right) + \frac{2}{s}e^{-3s}\right]$

(e) $f(t) = \begin{cases} \dfrac{3t}{2} & \text{if } 0 \le t < 2 \\ 6 - \dfrac{3t}{2} & \text{if } 2 \le t < 4 \end{cases} \qquad f(t+4) = f(t)$

Ans. $\mathcal{L}\{f(t)\} = \frac{3(1-e^{-2s})}{2s^2(1+e^{-2s})}$

(f) $f(t) = \begin{cases} 1 & \text{if } 0 \le t < 7 \\ -1 & \text{if } 7 \le t < 14 \end{cases} \qquad f(t+14) = f(t)$ **Ans.** $\mathcal{L}\{f(t)\} = \frac{e^{7s}-1}{s(e^{7s}+1)}$

(g) $f(t) = \begin{cases} 2 & \text{if } 0 \le t < 3 \\ -2 & \text{if } 3 \le t < 6 \end{cases} \qquad f(t+14) = f(t)$ **Ans.** $\mathcal{L}\{f(t)\} = \frac{2e^{3s}-2}{s(e^{3s}+1)}$

(h) $f(t) = 9t \qquad f(t+7) = f(t)$ **Ans.** $\mathcal{L}\{f(t)\} = \frac{9e^{7s}-63s-9}{s^2(e^{7s}-1)}$

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Inverse Laplace transform

Definition(Inverse Laplace transform)

If $f(t)$ has the Laplace transform $F(s)$, i.e., $\mathcal{L}\{f(t)\} = F(s)$, then the inverse Laplace transform of $F(s)$ is denoted by $\mathcal{L}^{-1}\{F(s)\}$ and is defined as

$$\mathcal{L}^{-1}\{F(s)\} = f(t)$$

Inverse Laplace transform

Definition(Inverse Laplace transform)

If $f(t)$ has the Laplace transform $F(s)$, i.e., $\mathcal{L}\{f(t)\} = F(s)$, then the inverse Laplace transform of $F(s)$ is denoted by $\mathcal{L}^{-1}\{F(s)\}$ and is defined as

$$\mathcal{L}^{-1}\{F(s)\} = f(t)$$

Theorem (Linearity).

The inverse Laplace transform is linear, i.e.,

$$\mathcal{L}^{-1}\{aF_1(s) + bF_2(s)\} = a\mathcal{L}^{-1}\{F_1(s)\} + b\mathcal{L}^{-1}\{F_2(s)\}$$

where a and b are arbitrary constants.

Theorem.

If $\mathcal{L}^{-1}\{F(s)\} = f(t)$, *then*

$$\mathcal{L}^{-1}\{F(s-a)\} = e^{at} f(t)$$

Theorem.

If $\mathcal{L}^{-1}\{F(s)\} = f(t)$, *then*

$$\mathcal{L}^{-1}\{F(s-a)\} = e^{at}f(t)$$

Theorem.

If $\mathcal{L}^{-1}\{F(s)\} = f(t)$, *then*

$$\mathcal{L}^{-1}\{e^{-as}F(s)\} = u_a(t)f(t-a) = \begin{cases} f(t-a) & t \geq a \\ 0 & t < a \end{cases}$$

Inverse Laplace Transform using partial fraction

We can use partial fraction to find the inverse Laplace transform of rational function. The first step is to factor the denominator as much as possible. Then for each term in the denominator we will use the following table to get a term or terms for our partial fraction decomposition.

Factor in denominator	**Term in partial fraction decomposition**
$ax+b$	$\frac{A}{ax+b}$
$(ax+b)^k$	$\frac{A_1}{ax+b}+\frac{A_2}{(ax+b)^2}+\cdots+\frac{A_k}{(ax+b)^k}$
ax^2+bx+c	$\frac{Ax+B}{ax^2+bx+c}$
$(ax^2+bx+c)^k$	$\frac{A_1x+B_1}{ax^2+bx+c}+\frac{A_2x+B_2}{(ax^2+bx+c)^2}+\cdots+\frac{A_kx+B_k}{(ax^2+bx+c)^k}$

Example

Find the inverse Laplace transform of $F(s) = \dfrac{s-2}{s^2+4s+13}$.

Example

Find the inverse Laplace transform of $F(s) = \dfrac{s-2}{s^2+4s+13}$.

Solution.

$$
\begin{aligned}
F(s) = \frac{s-2}{s^2+4s+13} &= \frac{s-2}{(s+2)^2+9} \\
&= \frac{s+2}{(s+2)^2+3^2} - \frac{4}{(s+2)^2+3^2} \\
&= \frac{s+2}{(s+2)^2+3^2} - \frac{4}{3}\,\frac{3}{(s+2)^2+3^2}
\end{aligned}
$$

taking inverse Laplace transform on both side

$$
\mathcal{L}^{-1}\{F(s)\} = \mathcal{L}^{-1}\left\{\frac{s+2}{(s+2)^2+3^2}\right\} - \frac{4}{3}\mathcal{L}^{-1}\left\{\frac{3}{(s+2)^2+3^2}\right\}
$$

$$
\mathcal{L}^{-1}\{F(s)\} = e^{-2t}\cos 3t - \frac{4}{3}e^{-2t}\sin 3t
$$

Example

Find the inverse Laplace transform of $\dfrac{s^2}{(s+3)^3}$.

Example

Find the inverse Laplace transform of $\dfrac{s^2}{(s+3)^3}$.

Solution.

Use the partial fractions to write

$$\frac{s^2}{(s+3)^3} = \frac{A}{s+3} + \frac{B}{(s+3)^2} + \frac{C}{(s+3)^3}$$
$$\implies s^2 = A(s+3)^2 + B(s+3) + C$$
$$\implies s^2 = As^2 + (6A+B)S + (9A+3B+C)$$

Comparing $A = 1$, $B = -6$ *and* $C = 9$. *Thus*

$$\frac{s^2}{(s+3)^3} = \frac{1}{s+3} - \frac{6}{(s+3)^2} + \frac{9}{(s+3)^3}$$

Taking inverse Laplace transform on both sides

$$\mathcal{L}^{-1}\left\{\frac{s^2}{(s+3)^3}\right\} = \mathcal{L}^{-1}\left\{\frac{1}{s+3}\right\} - \mathcal{L}^{-1}\left\{\frac{6}{(s+3)^2}\right\} + \mathcal{L}^{-1}\left\{\frac{9}{(s+3)^3}\right\}$$
$$= e^{-3t} - 6te^{-3t} + \frac{9}{2}t^2e^{-3t}$$

Example

Find the inverse transform of each of the following

$$F(s) = \frac{2-5s}{(s-6)(s^2+11)}$$

Solution

The partial fraction decomposition of $F(s)$ is given by

$$F(s) = \frac{A}{s-6} + \frac{Bs+C}{s^2+11}$$

Setting numerators equal gives,

$$2-5s = A\left(s^2+11\right) + (Bs+C)(s-6)$$

completely multiply out the right side and collect like terms

$$\begin{aligned} 2-5s &= A\left(s^2+11\right) + (Bs+C)(s-6) \\ &= As^2 + 11A + Bs^2 - 6Bs + Cs - 6C \\ &= (A+B)s^2 + (-6B+C)s + 11A - 6C \end{aligned}$$

Solution cont. . .

Setting coefficients equal gives the following system of equations that can be solved.

$$\left.\begin{array}{rr} s^2: & A+B=0 \\ s^1: & -6B+C=-5 \\ s^0: & 11A-6C=2 \end{array}\right\} \quad \Rightarrow \quad A=-\frac{28}{47}, \quad B=\frac{28}{47}, \quad C=-\frac{67}{47}$$

The partial fraction decomposition is then,

$$\begin{aligned} F(s) &= \frac{1}{47}\left(-\frac{28}{s-6}+\frac{28s-67}{s^2+11}\right) \\ &= \frac{1}{47}\left(-\frac{28}{s-6}+\frac{28s}{s^2+11}-\frac{67\frac{\sqrt{11}}{\sqrt{11}}}{s^2+11}\right) \end{aligned}$$

The inverse transform is then.

$$f(t)=\frac{1}{47}\left(-28\mathbf{e}^{6t}+28\cos\left(\sqrt{11}t\right)-\frac{67}{\sqrt{11}}\sin\left(\sqrt{11}t\right)\right)$$

Example

Find $\mathcal{L}^{-1}\left\{\ln\left(\frac{s-a}{s-b}\right)\right\}$

Example

Find $\mathcal{L}^{-1}\left\{\ln\left(\frac{s-a}{s-b}\right)\right\}$

Solution.

Let $F(s) = \ln\left(\frac{s-a}{s-b}\right) = \ln(s-a) - \ln(s-b)$. *Then*

$$\frac{d}{ds}F(s) = \frac{1}{s-a} - \frac{1}{s-b}$$

$$\implies \mathcal{L}^{-1}\left\{\frac{d}{ds}F(s)\right\} = \mathcal{L}^{-1}\left\{\frac{1}{s-a}\right\} - \mathcal{L}^{-1}\left\{\frac{1}{s-b}\right\}$$

$$\implies -t\mathcal{L}^{-1}\{F(s)\} = e^{at} - e^{bt}$$

$$\implies \mathcal{L}^{-1}\{F(s)\} = \frac{e^{bt} - e^{at}}{t}$$

Example

Find $\mathcal{L}^{-1}\left\{\ln \frac{s^2+a^2}{\sqrt{s+b}}\right\}$.

Example

Find $\mathcal{L}^{-1}\left\{\ln\frac{s^2+a^2}{\sqrt{s+b}}\right\}$.

Solution.

Let $F(s)=\ln\frac{s^2+a^2}{\sqrt{s+b}}=\ln(s^2+a^2)-\frac{1}{2}\ln(s+b)$. *Then*

$$F'(s)=\frac{2s}{s^2+a^2}-\frac{1}{2(s+b)}$$

$$\Longrightarrow \mathcal{L}^{-1}\{F'(s)\}=2\mathcal{L}^{-1}\left\{\frac{s}{s^2+a^2}\right\}-\frac{1}{2}\mathcal{L}^{-1}\left\{\frac{1}{s+b}\right\}$$

$$\Longrightarrow -t\mathcal{L}^{-1}\{F(s)\}=2\cos at-\frac{1}{2}e^{-bt}$$

$$\Longrightarrow \mathcal{L}^{-1}\{F(s)\}=\frac{e^{-bt}}{2t}-\frac{2\cos at}{t}$$

Exercise 6:

Find the inverse transform of each of the following.

(a) $F(s) = \frac{6}{s} - \frac{1}{s-8} + \frac{4}{s-3}$ **Ans.** $f(t) = 6 - \mathbf{e}^{8t} + 4\mathbf{e}^{3t}$

(b) $H(s) = \frac{19}{s+2} - \frac{1}{3s-5} + \frac{7}{s^5}$ **Ans.** $h(t) = 19\mathbf{e}^{-2t} - \frac{1}{3}\mathbf{e}^{\frac{5t}{3}} + \frac{7}{24}t^4$

(c) $F(s) = \frac{6s}{s^2+25} + \frac{3}{s^2+25}$ **Ans.** $f(t) = 6\cos(5t) + \frac{3}{5}\sin(5t)$

(d) $G(s) = \frac{8}{3s^2+12} + \frac{3}{s^2-49}$ **Ans.** $g(t) = \frac{4}{3}\sin(2t) + \frac{3}{7}\sinh(7t)$

Exercise 7:

Find the inverse transform of each of the following.

(a) $F(s) = \frac{6s-5}{s^2+7}$ **Ans.** $f(t) = 6\cos\left(\sqrt{7}t\right) - \frac{5}{\sqrt{7}}\sin\left(\sqrt{7}t\right)$

(b) $F(s) = \frac{1-3s}{s^2+8s+21}$ **Ans.** $f(t) = -3\mathbf{e}^{-4t}\cos\left(\sqrt{5}t\right) + \frac{13}{\sqrt{5}}\mathbf{e}^{-4t}\sin\left(\sqrt{5}t\right)$

(c) $G(s) = \frac{3s-2}{2s^2-6s-2}$

Ans. $g(t) = \frac{1}{2}\left(3\mathbf{e}^{\frac{3t}{2}}\cosh\left(\frac{\sqrt{13}}{2}t\right) + \frac{5}{\sqrt{13}}\mathbf{e}^{\frac{3t}{2}}\sinh\left(\frac{\sqrt{13}}{2}t\right)\right)$

(d) $H(s) = \frac{s+7}{s^2-3s-10}$ **Ans.** $h(t) = -\frac{5}{7}\mathbf{e}^{-2t} + \frac{12}{7}\mathbf{e}^{5t}$

Exercise 8:

Find the inverse transform of each of the following.

(a) $G(s) = \dfrac{86s - 78}{(s+3)(s-4)(5s-1)}$

Ans. $g(t) = -3\mathbf{e}^{-3t} + 2\mathbf{e}^{4t} + \mathbf{e}^{\frac{t}{5}}$

(b) $F(s) = \dfrac{2 - 5s}{(s-6)(s^2+11)}$

Ans. $f(t) = \frac{1}{47}\left(-28\mathbf{e}^{6t} + 28\cos\left(\sqrt{11}t\right) - \frac{67}{\sqrt{11}}\sin\left(\sqrt{11}t\right)\right)$

(c) $G(s) = \dfrac{25}{s^3(s^2+4s+5)}$

Ans. $g(t) = \frac{1}{5}\left(11 - 20t + \frac{25}{2}t^2 - 11\mathbf{e}^{-2t}\cos(t) - 2\mathbf{e}^{-2t}\sin(t)\right)$

Exercise 9:

Find the inverse Laplace transform of each of the following.

(a) $H(s) = \dfrac{s\mathbf{e}^{-4s}}{(3s+2)(s-2)}$

Ans.

$h(t) = u_4(t)f(t-4) = \begin{cases} f(t-4) & t \geq 4 \\ 0 & t < 4 \end{cases}$, **where** $f(t) = \frac{1}{12}\mathbf{e}^{-\frac{2t}{3}} + \frac{1}{4}\mathbf{e}^{2t}$

(b) $G(s) = \dfrac{5\mathbf{e}^{-6s} - 3\mathbf{e}^{-11s}}{(s+2)(s^2+9)}$

Ans. $g(t) = 5u_6(t) f(t-6) - 3u_{11}(t) f(t-11)$,
where $f(t) = \frac{1}{13}\left(\mathbf{e}^{-2t} - \cos(3t) + \frac{2}{3}\sin(3t)\right)$

(c) $F(s) = \dfrac{4s + \mathbf{e}^{-s}}{(s-1)(s+2)}$

Ans. $f(t) = \frac{4}{3}\mathbf{e}^{t} + \frac{8}{3}\mathbf{e}^{-2t} + u_1(t) h(t-1)$, **where** $h(t) = \frac{1}{3}\mathbf{e}^{t} - \frac{1}{3}\mathbf{e}^{-2t}$

(d) $G(s) = \dfrac{3s + 8\mathbf{e}^{-20s} - 2s\mathbf{e}^{-3s} + 6\mathbf{e}^{-7s}}{s^2(s+3)}$

Ans. $g(t) = 3f(t) - 2u_3(t) f(t-3) + 8u_{20}(t) h(t-20) + 6u_7(t) h(t-7)$
where $f(t) = \frac{1}{3} - \frac{1}{3}\mathbf{e}^{-3t}$, **and** $h(t) = -\frac{1}{9} + \frac{1}{3}t + \frac{1}{9}\mathbf{e}^{-3t}$

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Definition (Convolution)

The convolution of the two given functions $f(t)$ and $g(t)$ is written as $f * g$ and is defined as

$$f * g = \int_0^t f(u)g(t-u)du.$$

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Note

The commutative, associative and distributive properties hold true.

Definition (Convolution)

The convolution of the two given functions $f(t)$ and $g(t)$ is written as $f * g$ and is defined as

$$f * g = \int_0^t f(u)g(t-u)du.$$

Note

The commutative, associative and distributive properties hold true.

Example

Find the convolution of t and e^{at}.

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Note

The commutative, associative and distributive properties hold true.

Example

Find the convolution of t and e^{at}.

Solution.

$$t * e^{at} = \int_0^t ue^{a(t-u)}du = e^{at}\int_0^t ue^{-au}$$
$$= \frac{1}{a^2}(e^{at} - at - 1)$$

Example

Find the convolution of $\sin at$ and $\sin at$.

Example

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Solution.

$$\begin{aligned}
\sin at * \sin at &= \int_0^t \sin au \sin a(t-u)du \\
&= \frac{1}{2}\int_0^2 2\sin au \sin a(t-u)du \\
&= \frac{1}{2}\int_0^t [\cos(2au-at) - \cos at]du \\
&= \frac{1}{2}\left[\frac{\sin}{(2au-at)}2a - u\cos at\right]_0^t \\
&= \frac{\sin at - at\cos at}{2a}
\end{aligned}$$

Convolution theorem

Theorem (Convolution theorem).

If $f(t)$ and $g(t)$ are two functions such that $\mathcal{L}^{-1}\{F(s)\} = f(t)$ and $\mathcal{L}^{-1}\{G(s)\} = g(t)$, then

$$\mathcal{L}^{-1}\{F(s)G(s)\} = f * g = \int_0^t f(u)g(t-u)du$$

where $*$ is the convolution operator.

Example

Find $\mathcal{L}^{-1}\left\{\frac{s}{(s^2+1)(s-1)}\right\}$.

Example

Find $\mathcal{L}^{-1}\left\{\frac{s}{(s^2+1)(s-1)}\right\}$.

Solution.

We can easily observe that the inverse Laplace transforms of $F(s)=\frac{s}{s^2+1}$ *and* $G(s)=\frac{1}{s-1}$ *are* $f(t)=\cos t$ *and* $g(t)=e^{2t}$, *respectively.*

In order to find the inverse Laplace transform, we can use the convolution theorem

$$\mathcal{L}^{-1}\{F(s)G(s)\}=\int_0^t f(u)g(t-u)du$$

$$\implies \mathcal{L}^{-1}\left\{\frac{s}{(s^2+1)(s-1)}\right\}=\int_0^t \cos u\ e^{2(t-u)}du$$

$$=e^{2t}\int_0^t \cos u\ e^{-2u}du$$

$$=e^{2t}\left(\frac{e^{-2t}\sin t-2e^{-2t}\cos t+2}{5}\right)$$

$$=\frac{2}{5}e^{2t}+\frac{1}{5}(\sin t-2\cos t)$$

Example

Find $\mathcal{L}^{-1}\left[\frac{1}{(s-1)(s-2)}\right]$.

Example

Find $\mathcal{L}^{-1}\left[\frac{1}{(s-1)(s-2)}\right]$.

Solution

We note that this is a product of two functions

$$Y(s) = \frac{1}{(s-1)(s-2)} = \frac{1}{s-1}\frac{1}{s-2} = F(s)G(s)$$

We know the inverse transforms of the factors: $f(t) = e^t$ and $g(t) = e^{2t}$. Using the Convolution Theorem, we find $y(t) = (f * g)(t)$. We compute the convolution:

$$\begin{aligned}
\mathcal{L}^{-1}\left[\frac{1}{(s-1)(s-2)}\right] &= \int_0^t f(u)g(t-u)du \\
&= \int_0^t e^u e^{2(t-u)}du \\
&= e^{2t}\int_0^t e^{-u}du \\
&= e^{2t}\left[-e^t + 1\right] = e^{2t} - e^t.
\end{aligned}$$

Exercise 10:

Use the Convolution theorem to find the inverse Laplace transform of the following

(a) $F(s) = \dfrac{1}{s^2(s^2+4)}$ **Hint.** $f(t) = \dfrac{1}{2}\displaystyle\int_0^t \tau \sin 2(t-\tau)\,d\tau$

(b) $F(s) = \dfrac{s}{(s+2)(s^2+9)}$ **Hint.** $f(t) = \displaystyle\int_0^t e^{-2\tau} \cos 3(t-\tau)\,d\tau$

(c) $F(s) = \dfrac{s}{(s^2+4)(s^2+9)}$ **Hint.** $f(t) = \dfrac{1}{2}\displaystyle\int_0^t \sin 2\tau \cos 3(t-\tau)\,d\tau$

(d) $F(s) = \dfrac{s}{(s^2+1)^2}$ **Hint.** $f(t) \displaystyle\int_0^t \cos\tau \sin(t-\tau)\,d\tau$

(e) $F(s) = \dfrac{1}{s(s-a)}$ **Hint.** $f(t) = \displaystyle\int_0^t e^{a\tau}\,d\tau$

(f) $F(s) = \dfrac{1}{(s+1)(s^2+2s+2)}$ **Hint.** $f(t) = e^{-t}\displaystyle\int_0^t \sin(t-\tau)\,d\tau$

(g) $F(s) = \dfrac{1}{(s+1)^2(s^2+4s+5)}$ **Hint.** $f(t) = e^{-2t}\displaystyle\int_0^t \tau e^{\tau} \sin(t-\tau)\,d\tau$

(h) $F(s) = \dfrac{1}{(s-1)^3(s+2)^2}$ **Hint.** $f(t) = \dfrac{e^{-2t}}{2}\displaystyle\int_0^t \tau^2(t-\tau)e^{3\tau}\,d\tau$

(i) $F(s) = \dfrac{s-1}{s^2(s^2-2s+2)}$ **Hint.** $f(t) = \displaystyle\int_0^t (t-\tau)e^{\tau}\cos\tau\,d\tau$

(j) $F(s) = \dfrac{s(s+3)}{(s^2+4)(s^2+6s+10)}$ **Hint.** $f(t) = \displaystyle\int_0^t e^{-3\tau}\cos\tau\cos 2(t-\tau)\,d\tau$

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5 Application of Laplace transform in solving ODEs

Application of Laplace transform in solving ODEs

We've spent the last three sections learning how to take Laplace transforms and how to take inverse Laplace transforms. These are going to be invaluable skills for the this section. We'll need the following result which we have already discussed in details in Section 1:

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Fact

If $f(t)$ is an n-times differentiable function of t, and $\mathcal{L}\{f(t)\} = F(s)$, then

$$\mathcal{L}\{f^{(n)}(t)\} = s^n F(s) - s^{n-1} f(0) - s^{n-2} f'(0) - \cdots - s f^{(n-2)}(0) - f^{(n-1)}(0)$$

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Let $Y(s) = \mathcal{L}\{y(t)\}$. Then, using above fact, we have

$$\mathcal{L}\{y'\} = sY(s) - y(0)$$

$$\mathcal{L}\{y''\} = s^2 Y(s) - sy(0) - y'(0) \qquad \text{and so on}$$

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For solving ODEs using Laplace transforms, we need to do following steps

- Take the Laplace transform of both sides of the given differential equation and then find the value of $Y(s)$.
- Now, take inverse Laplace transform of both the sides to get the solution $y(t)$ of given ODE.

Example

Find the solution of the following first order ODE

$$\frac{dy}{dt} + 3y = 0, \quad y(0) = 1$$

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$$\frac{dy}{dt} + 3y = 0, \quad y(0) = 1$$

Solution.

By taking Laplace transform of both sides of given ODE, we get

$$\mathcal{L}\left\{\frac{dy}{dt}\right\} + 3\mathcal{L}\{y\} = 0$$
$$\implies sY(s) - y(0) + 3Y(s) = 0$$
$$\implies (s+3)Y(s) - 1 = 0$$
$$\implies Y(s) = \frac{1}{s+3}$$

Taking inverse Laplace transform of both sides, yields

$$\mathcal{L}^{-1}\{Y(s)\} = \mathcal{L}^{-1}\left\{\frac{1}{s+3}\right\}$$
$$\implies y(t) = e^{-2t}$$

Example

Find the solution of the following second order ODE

$$\frac{d^2y}{dt^2} + y = t, \quad y(0) = 1, \quad y'(0) = -2$$

Example

Find the solution of the following second order ODE

$$\frac{d^2y}{dt^2} + y = t, \quad y(0) = 1, \quad y'(0) = -2$$

Solution.

By taking Laplace transform of both sides of given ODE, we get

$$\mathcal{L}\left\{\frac{d^2y}{dt^2}\right\} + \mathcal{L}\{y\} = \mathcal{L}\{t\}$$

$$\implies s^2Y(s) - sy(0) - y'(0) + Y(s) = \frac{1}{s^2}$$

$$\implies (s^2+1)Y(s) - s + 2 = \frac{1}{s^2}$$

$$\implies Y(s) = \frac{1}{s^2(s^2+1)} + \frac{s-2}{s^2+1}$$

$$\implies Y(s) = \frac{1}{s^2} + \frac{s}{s^2+1} - \frac{3}{s^2+1}$$

Taking inverse Laplace transform of both sides, yields

$$y(t) = t + \cos t - 3\sin t$$

Example

Solve the following IVP.

$$y'' - 10y' + 9y = 5t, \qquad y(0) = -1 \quad y'(0) = 2$$

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Solution.

By taking Laplace transform of both sides of given ODE, we get

$$\mathcal{L}\{y''\} - 10\mathcal{L}\{y'\} + 9\mathcal{L}\{y\} = \mathcal{L}\{5t\}$$

$$\implies s^2 Y(s) - sy(0) - y'(0) - 10(sY(s) - y(0)) + 9Y(s) = \frac{5}{s^2}$$

$$\implies s^2 Y(s) + s - 10sY(s) - 10 + 9Y(s) = \frac{5}{s^2}$$

$$\implies Y(s) = \frac{5}{s^2(s-9)(s-1)} + \frac{12-s}{(s-9)(s-1)}$$

$$\implies Y(s) = \frac{50}{81}\frac{1}{s} + \frac{5}{9}\frac{1}{s^2} + \frac{31}{81}\frac{1}{s-9} - \frac{2}{s-1}$$

Finally taking the inverse transform gives us the solution to the IVP.

$$y(t) = \frac{50}{81} + \frac{5}{9}t + \frac{31}{81}\mathbf{e}^{9t} - 2\mathbf{e}^{t}$$

Example

Solve the following IVP.

$$2y'' + 3y' - 2y = t\mathbf{e}^{-2t}, \qquad y(0) = 0 \quad y'(0) = -2$$

Example

Solve the following IVP.

$$2y'' + 3y' - 2y = t\mathbf{e}^{-2t}, \qquad y(0) = 0 \quad y'(0) = -2$$

Solution.

By taking Laplace transform of both sides of given ODE, we get

$$2\mathcal{L}\{y''\} + 3\mathcal{L}\{y'\} - 2\mathcal{L}\{y\} = \mathcal{L}\left\{te^{-2t}\right\}$$

$$2\left(s^2Y(s) - sy(0) - y'(0)\right) + 3\left(sY(s) - y(0)\right) - 2Y(s) = \frac{1}{(s+2)^2}$$

$$\implies \left(2s^2 + 3s - 2\right)Y(s) + 4 = \frac{1}{(s+2)^2}$$

$$\implies Y(s) = \frac{1}{(2s-1)(s+2)^3} - \frac{4}{(2s-1)(s+2)}$$

$$\implies Y(s) = \frac{1}{125}\left(\frac{-192}{2\left(s-\frac{1}{2}\right)} + \frac{96}{s+2} - \frac{10}{(s+2)^2} - \frac{25\frac{2!}{2!}}{(s+2)^3}\right)$$

Taking the inverse transform then gives,

$$y(t) = \frac{1}{125}\left(-96\mathbf{e}^{\frac{t}{2}} + 96\mathbf{e}^{-2t} - 10t\mathbf{e}^{-2t} - \frac{25}{2}t^2\mathbf{e}^{-2t}\right)$$

Example

Solve the following IVP.

$$y'' - 6y' + 15y = 2\sin(3t), \qquad y(0) = -1 \quad y'(0) = -4$$

Example

Solve the following IVP.

$$y'' - 6y' + 15y = 2\sin(3t), \qquad y(0) = -1 \quad y'(0) = -4$$

Solution.

By taking Laplace transform of both sides of given ODE, we get

$$s^2 Y(s) - sy(0) - y'(0) - 6(sY(s) - y(0)) + 15Y(s) = 2\frac{3}{s^2+9}$$

$$\implies (s^2 - 6s + 15)\, Y(s) + s - 2 = \frac{6}{s^2+9}$$

$$\implies Y(s) = \frac{-s^3 + 2s^2 - 9s + 24}{(s^2+9)(s^2 - 6s + 15)}$$

First let's get the partial fraction decomposition.

$$Y(s) = \frac{As + B}{s^2+9} + \frac{Cs + D}{s^2 - 6s + 15}$$

Now, setting numerators equal gives,

$$\begin{aligned} -s^3 + 2s^2 - 9s + 24 &= (As + B)(s^2 - 6s + 15) + (Cs + D)(s^2 + 9) \\ &= (A + C)s^3 + (-6A + B + D)s^2 + (15A - 6B + 9C)s + 15B + 9D \end{aligned}$$

Solution Cont...

Setting coefficients equal and solving for the constants gives,

$$\left.\begin{array}{rr} s^3: & A+C=-1 \\ s^2: & -6A+B+D=2 \\ s: & 15A-6B+9C=-9 \\ s^0: & 15B+9D=24 \end{array}\right\} \quad \Rightarrow \quad \begin{array}{ll} A=\frac{1}{10} & B=\frac{1}{10} \\ C=-\frac{11}{10} & D=\frac{5}{2} \end{array}$$

$$\begin{aligned} Y(s) &= \frac{1}{10}\left(\frac{s+1}{s^2+9}+\frac{-11s+25}{s^2-6s+15}\right) \\ &= \frac{1}{10}\left(\frac{s+1}{s^2+9}+\frac{-11(s-3+3)+25}{(s-3)^2+6}\right) \\ &= \frac{1}{10}\left(\frac{s}{s^2+9}+\frac{1\frac{3}{3}}{s^2+9}-\frac{11(s-3)}{(s-3)^2+6}-\frac{8\frac{\sqrt{6}}{\sqrt{6}}}{(s-3)^2+6}\right) \end{aligned}$$

Taking the inverse transform then gives,

$$y(t)=\frac{1}{10}\left(\cos(3t)+\frac{1}{3}\sin(3t)-11\mathbf{e}^{3t}\cos\left(\sqrt{6}t\right)-\frac{8}{\sqrt{6}}\mathbf{e}^{3t}\sin\left(\sqrt{6}t\right)\right)$$

Example

Solve $\frac{d^2x}{dt^2} + 9x = \cos 2t, x(0) = 1, x\left(\frac{\pi}{2}\right) - 1.$

Example

Solve $\frac{d^2x}{dt^2} + 9x = \cos 2t, x(0) = 1, x\left(\frac{\pi}{2}\right) - 1.$

Solution

Since $x'(0)$ is not given, we assume $x'(0) = a$. By taking Laplace transform of both sides of given ODE, we get

$$s^2X(s) - sx(0) - x'(0) + 9(sX(s) - x(0)) = \frac{s}{s^2+4}$$

$$\implies (s^2 9s)X(s) - s - a = \frac{s}{s^2+4}$$

$$\implies X(s) = \frac{s+a}{s^2+9} + \frac{s}{(s^2+4)(s^2+9)}$$

$$\implies X(s) = \frac{a}{s^2+9} + \frac{1}{5}\cdot\frac{s}{s^2+4} + \frac{4}{5}\cdot\frac{s}{s^2+9}$$

Taking inverse Laplace transform of both sides yields

$$x(t) = \frac{a}{3}\sin 3t + \frac{1}{5}\cos 2t + \frac{4}{5}\cos 3t$$

when $t = \frac{\pi}{2}$, $-1 = -\frac{a}{3} - \frac{1}{5} \implies a = \frac{12}{5}$. Therefore

$$x(t) = \frac{1}{5}[4\sin 3t + \cos 2t + 4\cos 3t]$$

Exercise 11:

Use the method of Laplace transforms to solve each of the following IVPs.

(a) $y' + 10y = t^2, \quad y(0) = 0$

Ans. $y = \frac{t^2}{100} - \frac{t}{50} + \frac{1}{500} - \frac{1}{500}e^{-10t}$

(b) $y' + 2y = t^{-t}, \quad y(0) = 2$

Ans. $y = te^{-t} - e^{-t} + 3e^{-2t}$

(c) $y' - 6y = 2\sin 3t, \quad y(0) = -1$

Ans. $y = \frac{-13}{15}e^{6t} - \frac{2}{15}\cos 3t - \frac{4}{15}\sin 3t$

(d) $y'' + 2y' = te^{-t}, \quad y(0) = 6 \quad y'(0) = -1$

Ans. $y = -te^{-t} + 6$

(e) $y'' + 4y' - 5y = 0,, \quad y(0) = 5 \quad y'(0) = -1$

Ans. $y = 4e^{t} + e^{-5t}$

(f) $y'' - 2y' + y = 2t - 3, \quad y(0) = 5 \quad y'(0) = 11$

Ans. $y = 4e^{t} + 5te^{t} + 2t + 1$

(g) $y'' + 8y' + 25y = 13e^{-2t}, \quad y(0) = -1 \quad y'(0) = 18$

Ans. $y = -2e^{-4t}\cos 3t + 4e^{-4t}\sin 3t + e^{-2t}$

(h) $y'' + 6y' + 34y = 0, \quad y(0) = -1 \quad y'(0) = 13$

Ans. $y = -e^{-3t}\cos 5t + 2e^{-3t}\sin 5t$

(i) $y'' + 4y = 8\cos 2t - 8e^{-2t}, \quad y(0) = -2 \quad y'(0) = 0$

Ans. $y = -\cos 2t - \sin 2t + 2t\sin 2t - e^{-2t}$

(j) $y'' - 4y' + 4y = 8t^2 - 16t + 4, \quad y(0) = 0 \quad y'(0) = 1$

Ans. $y = te^{2t} + 2t^2$